

Aim - Design and implement a program to simulate a Page Replacement Algorithm using FIFO (First-In-First-Out) in a virtual memory system. The program should load pages into memory frames as needed and replace the oldest page when a page fault occurs. Compute the total number of page faults and page hits for a given reference.

Code:

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005
GNU nano 8.7
#include <stdio.h>

int main() {
    int frames, n;

    printf("Enter number of frames: ");
    scanf("%d", &frames);

    printf("Enter number of pages: ");
    scanf("%d", &n);

    int pages[n];
    printf("Enter page reference string:\n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    int frame[frames];
    int front = 0;
    int page_faults = 0, hits = 0;

    // Initialize frames
    for(int i = 0; i < frames; i++) {
        frame[i] = -1;
    }

    for(int i = 0; i < n; i++) {
        int found = 0;

        // Check if page is already in frame
        for(int j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                found = 1;
                hits++;
                break;
            }
        }

        // If not found → Page Fault
        if(!found) {
            frame[front] = pages[i];
            front = (front + 1) % frames;
            page_faults++;
        }
    }

    printf("\nPage Hits = %d\n", hits);
    printf("Page Faults = %d\n", page_faults);

    return 0;
}
```

Output:

```
M /c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ cd /c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ nano fifo.c

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ gcc fifo.c -o fifo

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ./fifoo
-bash: ./fifoo: No such file or directory

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ./fifoo
Enter number of frames: 3
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5

Page Hits = 3
Page Faults = 9

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |
```

nce string, thereby analyzing memory utilization efficiency